

Literature review - Algal Blooms in the Baltic Sea: Environmental Drivers, Impacts, and Management Challenges

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Abstract

The Baltic Sea has become one of the world’s most bloom-affected marine environments, where dense cyanobacterial accumulations now occur annually across hundreds of thousands of square kilometers. This review synthesizes current understanding of algal bloom dynamics, examining environmental drivers, seasonal and spatial patterns, impacts, and monitoring approaches. Long-term phosphorus accumulation from decades of nutrient loading drives bloom intensity, while meteorological conditions determine annual occurrence. The Baltic’s characteristics as a shallow, semi-enclosed brackish sea favor nitrogen-fixing cyanobacteria, which contribute up to 792,000 tonnes of new nitrogen annually, perpetuating eutrophication. Satellite observations show expanding distributions, with blooms now extending into previously unaffected northern regions. Impacts extend beyond ecological disruption to affect fisheries through toxin contamination, tourism through beach closures, and public health through hepatotoxin exposure. Economic losses are substantial but not fully quantified. Monitoring has advanced through satellite sensing, autonomous systems, and ensemble forecasting, though coverage gaps remain. Machine learning shows promise but hasn’t been widely implemented. Climate projections suggest warmer, more stratified conditions will favor blooms, while freshening may alter species composition unpredictably. Despite challenges, coordinated nutrient reduction efforts under the Helsinki Commission’s Baltic Sea Action Plan, combined with advancing monitoring technologies, offer pathways for effective management requiring sustained regional commitment.

1 Introduction

The Baltic Sea is among the world’s most extensively studied marine systems. This shallow, semi-enclosed brackish sea is surrounded by a drainage basin home to approximately 84 million people [Munkes et al., 2021]. For over a century, human activities have significantly increased nutrient inputs to the sea. The effects of this nutrient loading are visible throughout the water column, on the seafloor, and increasingly at the surface, where dense, colorful algal blooms have become a defining characteristic of Baltic summers. These blooms are no longer rare, occasional events but annual occurrences that appear to be intensifying and expanding across many regions [Karlson et al., 2022]; [Viitasalo and Bonsdorff, 2022].

There are many different kinds of algal blooms in the Baltic Sea. The most ecologically and publicly visible are the huge summer surface accumulations of fiber-like, nitrogen-fixing cyanobacteria, mainly *Nodularia spumigena*, *Aphanizomenon sp.*, and *Dolichospermum spp.* They can cover hundreds of thousands of square kilometers of open water [Karlson et al., 2022]. These organisms have been part of the Baltic ecosystem for thousands of years [Munkes et al., 2021], but eutrophication has made their blooms much bigger and more common. Macroalgal blooms, especially those of the green algae *Cladophora glomerata*, cause localized but severe near-shore hypoxia in the Baltic’s shallower coastal areas [Gubelit and Berezina, 2010]. In addition to these primary forms, the Baltic and its neighboring northern European seas are home to a varied community of toxic phytoplankton, such as *Alexandrium* species, *Dinophysis* species, and *Pseudo-nitzschia* species, whose bloom dynamics, toxin profiles, and environmental change responses are altering due to climate pressure [Karlson et al., 2021]; [Tulatz et al., 2025].

Although the broad environmental drivers underlying these blooms are well established, little is known about how they interact and how important they are over different periods of time. Phosphorus is the key long-term driver of cyanobacterial blooms in the open Baltic Sea. A major source of this nutrient is the release from oxygen-depleted sediments on the seafloor - a process that has been linked over decades to the buildup of excess phosphorus in the water, the spread of low-oxygen zones, and the increasing frequency of surface blooms [Kahru et al., 2020]. From

year to year, physical and weather conditions, particularly sunlight and wind speed in July, control bloom timing and occurrence. Sea surface temperature (SST) emerges as a result of the same solar radiation conditions that cause blooms, rather than as a cause of them [Kahru et al., 2025]. This distinction between decadal nutrient-driven trends and meteorologically driven year-to-year variability is critical, as it fundamentally limits the ability of water chemistry alone to predict bloom occurrence in any given summer [Kahru et al., 2020]; [Munkes et al., 2021].

Another complicating factor to consider is the role of cyanobacteria in contributing to the eutrophication they respond to. The cyanobacteria of the Baltic Sea are estimated to add 434,000 to 792,000 tonnes of bioavailable nitrogen to the already over-fertilized environment through the process of biological nitrogen fixation each year. This contribution has the potential to be of similar magnitude to the nitrogen reduction goals set by the Helsinki Commission’s Baltic Sea Action Plan [Munkes et al., 2021]. This has resulted in a cycle where hypoxia leads to phosphorus release, which leads to cyanobacteria blooms, which leads to further oxygen consumption. The persistence of this cycle has been one of the primary challenges for the environmental governance of the Baltic Sea environment despite decades of internationally coordinated nutrient management [Munkes et al., 2021]; [Viitasalo and Bonsdorff, 2022].

The impacts of algal blooms stretch far beyond the ecological. When blooms are formed by toxic or otherwise harmful species, they are collectively referred to as Harmful Algal Blooms (HABs). This term encompasses a broad range of phytoplankton and cyanobacteria capable of producing toxins, depleting oxygen, or physically damaging aquatic organisms [Karlson et al., 2021]. In the Baltic Sea, HABs occur in several forms, which are ecologically and economically destructive. Cyanobacterial toxins, especially the hepatotoxin nodularin produced by *N. spumigena*, accumulate through food chains and pose risks to fish, mammals, and humans [Karlson et al., 2022]. In coastal areas, the degradation of drifting seaweed beds can deplete oxygen in the nearshore zones, killing off the small animals that live there [Gubelit and Berezina, 2010]. On a wider northern European scale, HABs have been associated with catastrophic losses in aquaculture, shellfish harvesting, and human health incidents, as well as massive fish kills due to toxic microscopic algae in Norwegian and Scandinavian waters [Karlson et al., 2021]. HABs pose a serious and growing threat to tourism income in Baltic coastal areas. Surveys of coastal tourists in southern Sweden show that algal blooms are widely known among visitors and cause negative reactions, leading many to cancel holidays or avoid affected destinations [Nilsson and Gössling, 2013]. Economic studies of HAB-impacted fishing communities have confirmed that the economic costs of fishery closures and reduced tourism activity far exceed the economic investment required for effective monitoring and early warning systems [Weir et al., 2022].

Tackling these challenges requires better tools for tracking and predicting algal blooms. Methods like satellite imagery, long-term water sampling, and automated sensors on ships all help scientists understand how blooms develop, but each only captures part of the picture, and combining them effectively is still a work in progress [Karlson et al., 2022]. By using multiple biogeochemical modelling approaches together, scientists have shown that it is possible to forecast algal blooms weeks or months in advance. However, verification is limited by observational data scarcity [Roiha et al., 2010]. More recently, machine learning approaches such as deep learning, random forests, and explainable artificial intelligence have demonstrated great promise for enhancing the accuracy of bloom prediction from environmental data sets and satellite imagery [Park et al., 2024]. At the same time, advances in molecular biology have revealed new opportunities for monitoring the transfer of toxic cyanobacterial genotypes from freshwater systems into the coastal waters of the Baltic [Fournier et al., 2025].

This review synthesizes the current state of knowledge regarding algal blooms in the Baltic Sea through five interlinked topics: the environmental drivers for algal bloom formation, the temporal and spatial patterns of algal blooms in different sub-basins of the Baltic Sea, the long-term trends in algal blooms with climate change, the ecological and socio-economic effects of algal blooms, and the changing landscape of monitoring, forecasting, and early warning systems. By drawing on evidence from observational studies, experiments, models, and socio-economic research, this review attempts to clarify areas of consensus, uncertainties, and research needs. With the continued warming, freshening, and nutrient enrichment of the Baltic Sea predicted in future climate change scenarios, the urgency for advancing our understanding of algal bloom dynamics has never been more critical.

2 The Baltic Sea as a Unique Environment for Algal Bloom Development

In order to understand why algal blooms are such a persistent and worsening problem in the Baltic Sea, it is crucial to first appreciate the unusual physical and ecological character of the sea itself. The Baltic is not a typical ocean environment. It is a shallow, semi-enclosed body of water in northern Europe, connected to the open North Sea only through the narrow and relatively shallow Danish Straits. This restricted water exchange means that nutrients, pollutants, and other substances entering the Baltic tend to accumulate within it rather than being diluted or flushed away. That is a fundamental feature, which makes the sea inherently vulnerable to the effects of human activity [Viitasalo and Bonsdorff, 2022].

The Baltic is also unusually brackish. Fed by numerous large rivers draining a wide land, and receiving more freshwater from precipitation than it loses to evaporation, the sea maintains a persistent salinity gradient from nearly fresh conditions in the northernmost Gulf of Bothnia to more saline, though still well below ocean levels, conditions in the south and west. This gradient defines which species can thrive where, and plays an important role in determining the composition and behavior of algal communities across the sea’s different regions [Munkes et al., 2021]; [Tulatz et al., 2025]. The contrast between lighter, fresher surface water and denser, saltier deep water creates strong layering during summer, trapping warm, nutrient-rich water near the surface and providing ideal conditions for algal growth [Viitasalo and Bonsdorff, 2022].

The region draining into the Baltic Sea is home to approximately 84 million people across nine countries, and the sea has absorbed the environmental consequences of their agricultural, industrial, and urban activities for well over a century [Munkes et al., 2021]. From the mid-twentieth century onward, intensive farming and inadequate wastewater treatment led to dramatically elevated levels of nitrogen and phosphorus, the two nutrients that most directly fuel algal growth, into Baltic rivers and, ultimately, into the sea itself. Although significant international efforts to reduce nutrient loading have been ongoing since the 1970s under the Helsinki Convention, and more recently through the Helsinki Commission’s Baltic Sea Action Plan, the ecological response has been disappointingly slow [Viitasalo and Bonsdorff, 2022]. The reason for this lies not only at the surface but on the seafloor as well.

A key and often underestimated feature of the Baltic’s eutrophication problem is the role of internal nutrient loading. Decades of excessive algal growth and the following sinking and decomposition of organic matter have stripped large areas of the Baltic seabed of oxygen, creating so-called ‘dead zones’ where little marine life can survive. Under those low-oxygen conditions, phosphorus that would otherwise remain locked in the seafloor is released back into the water, where it becomes available to fuel yet more algal growth at the surface [Munkes et al., 2021]; [Viitasalo and Bonsdorff, 2022]. This creates a self-reinforcing cycle, often described as the ‘vicious circle of eutrophication’, in which the oxygen depletion drives nutrient release, which drives more algal growth, which drives more oxygen depletion. This process means that even if all external nutrient inputs were stopped tomorrow, the Baltic would continue to carry a large internal nutrient debt that sustains bloom conditions for decades to come [Viitasalo and Bonsdorff, 2022].

Adding a further layer of complexity, the cyanobacteria that dominate Baltic summer blooms have a remarkable ability to draw nitrogen directly from the atmosphere through a process called nitrogen fixation - effectively creating their own fertilizer when dissolved nitrogen in the water runs low [Munkes et al., 2021]. This biological capability gives them a powerful competitive advantage over the phytoplankton during the nitrogen-depleted summer months, and means that reducing nitrogen inputs to the sea alone is unlikely to suppress their growth. Estimates suggest that cyanobacteria introduce between 434 000 and 792 000 tonnes of new nitrogen into the Baltic each year through fixation. This is an amount that may rival or even exceed the nitrogen reductions targeted by current international management plans [Munkes et al., 2021]. This additional nitrogen also feeds back into the eutrophication cycle, further complicating the path to ecological recovery.

The Baltic Sea presents a particularly difficult management problem. Its physical enclosure limits natural recovery, its long history of nutrient loading has created a persistent internal reservoir of pollutants, its unique salinity and layering of waters favor bloom-forming cyanobacteria, and the biological traits of those cyanobacteria allow them to partially bypass the effects of nutrient reduction efforts. Each of these characteristics, on its own, would make the Baltic more prone to algal blooms than a typical open-ocean environment. Together, they help explain why the Baltic Sea has become one of the most bloom-affected seas in the world, and why understanding the drivers, dynamics, and consequences of those blooms is a matter of urgent scientific and policy relevance [Munkes et al., 2021]; [Aleksandrov, 2010]; [Viitasalo and Bonsdorff, 2022].

3 Environmental Drivers of Bloom Formation

Algal blooms do not appear randomly. They arise when a specific combination of environmental conditions aligns to give bloom-forming species a competitive advantage over other organisms in the water. In the Baltic Sea, this combination involves nutrients accumulated over decades, physical conditions that concentrate algae near the surface, weather patterns that calm the water at just the right moment, and increasingly, the influence of a warming climate. Understanding which of these factors matters most, and at what timescale, has been one of the central points of Baltic Sea research for the past several decades. What has emerged is a picture of bloom formation controlled by different drivers operating at different scales: long-term nutrient conditions set the stage, while short-term weather and physical conditions determine whether and when the performance actually begins.

3.1 Nutrient Loading and the Role of Phosphorus

At the broadest timescale, the most important driver of cyanobacterial bloom intensity in the Baltic Sea is the availability of phosphorus. In the Baltic, phosphorus serves as the primary currency for algal growth; when it is abundant, blooms are typically larger and more intense; when it is scarce, they are more limited. For periods of five to twenty years, researchers have found a strong link between the number of cyanobacterial surface accumulations in the central Baltic, the amount of phosphorus in the water and the size of oxygen-depleted zones on the seafloor [Kahru et al., 2020]. This long-term relationship shows how decades of agricultural drainage, urban wastewater discharge, and industrial pollution have steadily introduced nutrients to the Baltic Sea that the ecosystem can't manage.

What makes the phosphorus problem particularly hard to solve is that a large and growing proportion of it no longer comes from outside the sea at all. As described in the previous section, low-oxygen conditions on the Baltic seafloor cause phosphorus to be continuously released from the sediments back into the water. Studies of Baltic coastal lagoons, including the Curonian Lagoon on the Lithuanian and Russian coast, have illustrated this persistence in clear terms: cyanobacterial blooms persisted at the same intensity or higher in the 1990s, despite a three- to four-fold decrease in fertilizer use and industrial activity [Aleksandrov, 2010]. The phosphorus reservoir that has accumulated in the sediment during the previous decades continued to support blooms long after the external inputs that had caused them has been eliminated. This finding carries an important and disturbing implications for Baltic Sea management: lowering nutrient imports is crucial, but the ecological benefits may take much longer to manifest than policymakers anticipate.

3.2 Physical and Meteorological Triggers

While nutrients provide the raw material for bloom growth, it is the physical environment, particularly weather conditions, that determines whether a bloom actually forms in any given summer. A long-held belief that cyanobacterial blooms are caused by warmer sea surface temperatures has been called into question by research based on more than 20 years of satellite observations over 36 regions of the Baltic Sea. [Kahru et al., 2025]. The evidence actually suggests otherwise. Sea surface temperature doesn't seem to affect when blooms start; instead, the water's temperature at that moment reflects the same meteorological factors that initially caused the bloom, especially the amount of incoming solar radiation and the lack of strong winds [Kahru et al., 2025].

The mechanism works roughly as follows. When the sun is shining brightly and there is no wind, the surface of the sea warms up quickly and the water becomes stratified. This means that the warm, lighter surface water stays above the cooler, denser water below, with little mixing between the two layers. Cyanobacteria love this calm, well-lit, layered environment. They can regulate their position in the water to remain in the sunlit surface layer, while other phytoplankton species, which depend on mixing to bring nutrients up from depth, are disadvantaged. The result is that a period of sunny, windless summer weather can quickly shift the competitive balance in favor of cyanobacteria and trigger a bloom within days [Kahru et al., 2025]; [Roiha et al., 2010].

However, it is important to note that a bloom does not always happen, even when all of the already mentioned conditions are met [Kahru et al., 2025]. This suggests that biological factors such as food-web dynamics and existing population structures also play a role, one not yet fully captured in current monitoring or modeling approaches. One of the most significant open questions in Baltic bloom science is the difference between the conditions required for a bloom and those sufficient to ensure one. This gap is a major factor in the difficulty of accurately predicting blooms from year to year.

The physical drivers of macroalgal blooms function somewhat differently in coastal areas. Storm activity and water level fluctuations are important controlling factors for species such as *Cladophora glomerata* in the Neva estuary on the eastern coast of the Gulf of Finland. This is not because they stop blooms, but rather because they physically remove attached algal mats from the seafloor and move them into shallower nearshore areas where they gather, drift, and eventually decompose [Gubelit and Berezina, 2010]. Severe oxygen depletion in shallow coastal waters is the most direct ecological harm caused by this decomposition process rather than bloom itself.

3.3 Temperature, Salinity, and the Influence of Climate Change

Beyond their role as a short-term trigger, temperature and salinity are important in determining which species are able to bloom in the Baltic as all, and how this may change as the climate continue to warm. Only species that can withstand this intermediate and fluctuating salinity regime can survive and flourish in the Baltic due to its distinct brackish character. The Baltic is expected to gradually become fresher over the next few decades due to climate change and an increase in freshwater inputs from rivers and precipitation, which will benefit some species and harm others [Viitasalo and Bonsdorff, 2022].

Laboratory experiments on the toxic species *Alexandrium ostenfeldii* from the Baltic and the Arctic seas are an example of what the consequences of these changes might look like biologically. Baltic strains have shown significant variability in their ability to survive high temperature conditions, with some of them growing even at 27°C while others cannot survive past 25°C. This shows that different populations might have vastly different responses to the warming trend [Tulatz et al., 2025]. The populations in the Arctic, on the other hand, might have an advantage if the temperature rises. This could cause them to spread into the Baltic and bloom longer. The freshening of the waters in the western Baltic might have negative consequences for some of the already established HAB species in the area, even as the warming opens new opportunities elsewhere [Tulatz et al., 2025]. These findings highlight that the response of harmful algal communities to climate change is not consistent and it varies by species, region, and by balance between warming and freshening in any given area.

More broadly, climate projections for the Baltic Sea under both moderate and high emission scenarios suggest warmer surface waters, stronger layering of the water column, and an earlier start to summer-like conditions. Together, these changes are expected to lengthen the period during which cyanobacterial blooms can develop and allow them to spread further north [Viitasalo and Bonsdorff, 2022]; [Karlson et al., 2022]. Satellite observation already show increasing surface accumulations of cyanobacteria in the Bothnian Sea, the northernmost major basin of the Baltic, a trend linked to both rising temperatures and changes in nutrient conditions [Karlson et al., 2022].

4 Seasonal Cycles and Geographic Distribution of Algal Blooms

Algal blooms are not random events that occur unpredictably throughout the calendar and the map. Instead, they have a clear seasonal pattern that relates to the annual cycle of light, temperature, and nutrient supplies, and they have a clear spatial pattern. Their concentration is higher in certain areas of the sea that are more susceptible to the conditions that favor the development of the blooms. The patterns of the blooms are not static, and the recent decades have seen changes in the timing, magnitude, and distribution of the blooms that relate both to the legacy of the long period of eutrophication and the early signs of the impact of climate change.

4.1 The Seasonal Cycle of Cyanobacterial Blooms

The Baltic Sea has a well-defined yearly cycle of biological activity, reflecting the strong seasonal contrast between daylight and temperatures at high latitudes. In the spring, the first significant increase in phytoplankton growth, called the spring bloom, is caused by the increase in daylight and the upwelling of nutrients that had accumulated in deeper waters during the previous winter. This spring bloom is dominated primarily by microscopic algae and other non-harmful phytoplankton, and it rapidly consumes much of the dissolved nitrogen in the surface water. In early summer, the nitrogen supply is exhausted in the upper water layers, and the conditions begin to favor the growth of cyanobacteria, the only phytoplankton that can compensate for the nitrogen deficiency by using nitrogen gas dissolved in the air [Munkes et al., 2021].

From late June onwards, with the increase in temperature of the surface waters and the stratification of the water column in response to the long days and calm conditions, the cyanobacterial blooms start to form in the open waters of the Baltic Sea. Generally, the cyanobacterial blooms reach their maximum intensity in July and August, with the surface water temperatures at their peak, the days being long enough for intense solar radiation and with wind conditions most likely to be calm for long periods of time [Karlson et al., 2022]. During this period, the accumulations of cyanobacterial blooms on the surface can be enormous, with hundreds of thousands of square kilometers of bloom extend recorded by satellites in the most intense years, with the largest recorded cases observed in 2005, 2008, and 2018 [Karlson et al., 2022]. By September, with the decreasing water temperatures and the days becoming shorter, the cyanobacterial bloom season comes to an end, though the organic matter left behind by decomposing blooms continues to influence oxygen conditions on the seafloor well into the autumn and winter months.

For the shallower coastal areas of the Baltic, the annual pattern of the macroalgal blooms such as *Cladophora glomerata* has a slightly different pattern, where the growth of these species does not have a single peak in the summer, but rather two separate ones. One of them occurs in July, and the other one in September, indicating annual pattern of growth, storm-driven detachment, and regrowth [Gubelit and Berezina, 2010]. Although the ecological impact of the macroalgal blooms, such as the oxygen depletion of the near-shore environment of the Baltic, does not have a single annual window, the impact of the individual events varies depending on the intensity of the storms and the water-level fluctuations of the year.

4.2 Spatial Patterns Across the Baltic Regions

The Baltic Sea is not an uniform body of water, but rather a number of basins that are connected and have different physical, chemical, and biological properties. This is evidenced by the spatial distribution of algal blooms, which varies from region to region and has been altered over time in response to changes in the environmental conditions.

The Northern Baltic Proper, which is the large central region of the Baltic Sea extending between Sweden, Finland, and the Baltic states, has historically been the most frequently impacted area for surface accumulations of cyanobacteria. Satellite data from 2002 to 2020 confirms this as the zone with the highest occurrence of near-surface blooms [Karlson et al., 2022]. This is likely the result of the combination of factors that are particularly strong in this basin, including sufficient phosphorus availability, warm and often calm summer conditions, and the depth and temperature layers that are especially favorable for floating cyanobacteria. However, some years, the most intense surface blooms have occurred in the Southern Baltic Proper, nearer Poland and Germany, which is an example of the significant year-to-year variation that makes it so difficult to predict the geography of the most intense surface blooms of any given year [Karlson et al., 2022].

Perhaps the most important spatial pattern revealed in the data from the past few decades is the extension of the range of cyanobacterial blooms further north into the Bothnian Gulf. The Bothnian Gulf area has been considered a relatively pristine area, essentially untouched by the cyanobacteria bloom issue that have been so widespread in the regions to the south. Satellite data now reveal a clear increase in near-surface bloom accumulation in this region, with blooms starting later in the season than those in the Baltic Proper but becoming more frequent and extensive over time [Karlson et al., 2022]. The likely cause is the flow of phosphorus-rich water from the central Baltic Proper Sea via the Åland Sea, a phenomenon believed to be related to the severe bottom-water hypoxia in the central region, where the nutrient-rich water is carried close enough to the boundary to spill northward [Viitasalo and Bonsdorff, 2022]. This represents a worrying trend: a region that was previously protected from the worst effects of Baltic eutrophication is now showing signs of the same bloom patterns that have long characterized the more heavily affected southern areas.

4.3 Annual Variation and Prediction Difficulties

A significant variability from year to year has been documented which makes predicting the intensity of any individual summer’s bloom exceptionally difficult. The distinction between strong bloom years such as 2005, 2008, and 2018, and relatively weak bloom years in between, cannot be attributed to nutrient conditions, since the chemical environment in the Baltic changes relatively slowly, and yet the strength and extend of the blooms can vary greatly [Kahru et al., 2020]. As discussed in the previous section, the meteorological conditions in a given summer, particularly the amount of solar radiation and wind, are the main factors that determine the differences in the yearly development of the blooms [Kahru et al., 2020]; [Kahru et al., 2025].

Such a level of weather dependence means that there is a limited degree of accuracy that can be achieved in bloom forecasting. For example, a summer characterized by constant calm weather and sunshine is likely to have blooms of higher intensity than a cool and windy summer, despite the state of nutrient concentration in the water. However, seasonal weather patterns are difficult to predict beyond a few weeks in advance. Advanced forecasting techniques, in which several models of both ocean and atmosphere are run simultaneously in order to generate a range of possible outcomes instead of a single prediction, have shown promise in capturing this uncertainty and communicating it in a form useful for warning systems [Roiha et al., 2010]. However, validation of these techniques is limited by the relatively low amount of available data in the Baltic Sea during bloom periods. This is a gap that more extensive monitoring networks could help to close.

Furthermore, statistical analysis carried out in 36 areas of the Baltic during a period of 25 years using satellite data has also shown that bloom initiation is not instantaneous across the entire sea but varies in time from one area to another depending on local conditions of solar radiation, wind, and water column structure [Kahru et al., 2025]. This adds another dimension of complexity to bloom prediction and emphasizes the point that the Baltic Sea should not be treated as a single entity but as a patchwork of sub-environmental zones responding to common overall drivers in different ways.

5 Ecological and Socio-Economic Impacts

The importance of algal blooms in the Baltic Sea stretches far beyond the scientific. When algal blooms happen, it creates a chain of events that has far-reaching effects on the ecosystem, the economy, human health, and the lives of the millions of people living in, working on, or visiting the shores of the Baltic Sea. Understanding the full scope of these consequences is essential not only for understanding why bloom research matters, but for arguing why investing in monitoring, early warning and management is urgently needed. This section is focused on examining the damage that algal blooms cause to the environment, their impacts on fishing and aquaculture in the Baltic Sea, and their increasingly well-documented impacts on coastal tourism and recreation.

5.1 Ecological Consequences

The most direct ecological consequence of cyanobacterial blooms in the Baltic Sea is related to their production and transport of toxins. Several of the most important bloom-forming cyanobacteria in the Baltic Sea are known for their ability to produce potent toxins that can cause damage to a wide range of organisms. Perhaps the most significant of these is nodularin, a liver toxin produced by *Nodularia spumigena*, one of the most important bloom-forming cyanobacteria in the open Baltic Sea. However, nodularin is not merely stored in the water column where it is produced but is accumulated step-wise in the food web, building up in small organisms, fish, and ultimately in the larger organisms that consume them, including marine mammals and seabirds [Karlson et al., 2022]. Dogs that drink infected water during bloom events have been shown to suffer acute liver and kidney failure and die from nodularin consumption [Karlson et al., 2022]. For humans, direct contact with dense bloom water can provoke skin irritations and rash, although the long-term health implications of human exposure to low levels of the toxin through contaminated fish are of continuing scientific concern.

Apart from toxicity, impacts of cyanobacterial blooms on the overall ecology of the Baltic Sea can be seen in several ways. When a bloom finally dies off and sinks to the bottom of the sea, the decomposition of the accumulated organic matter consumes large quantities of dissolved oxygen. This extra oxygen demand can make the already low-oxygen conditions in the Baltic’s deeper waters even worse, making them completely anoxic. This can have serious effects on fish, small aquatic animals, and the seafloor communities that depend on oxygenated conditions to survive [Munkes et al., 2021]. During bloom periods, dense surface blooms block the light for other phytoplankton, underwater plants, and the animals that depend on them, effectively reducing the overall biodiversity present in the water column during bloom periods.

In the shallower waters, the impact of macroalgal blooms is just as severe, although on a smaller spatial scale. In the Neva estuary on the east coast of the Gulf of Finland, detailed field measurements were carried out during a macroalgal bloom and its subsequent decomposition, and in the affected regions a near-total collapse of seafloor animal life was documented. The levels of oxygen in the water, where drifting masses of decaying *Cladophora glomerata* macroalgae were found, fell to as low as 5% of normal levels, well below the level that is normally survivable for

small marine animals. This resulted in the mass death of these important seafloor creatures that are a key component of the food web in the region [Gubelit and Berezina, 2010].

5.2 Impacts on Fisheries and Aquaculture

The effect of algal blooms on fisheries and aquaculture represents some of the most economically evident consequences of the HAB problem in the Baltic Sea and broader northern European region. Blooms harm fish and shellfish in multiple ways: they cause direct mortality through toxic or oxygen-depleted water, contaminate both wild-caught and farmed shellfish with toxins unsafe for human consumption, and trigger reputational and market consequences.

In the wider northern European context, the financial losses due to HAB-related incidents in aquaculture have been substantial. Blooms of the fish-killing *Chrysochromulina leadbeateri* in northern Norway led to the loss of over 8.2 million farmed salmon in a single year in 2019, with a total loss of over 100 million US dollars [Karlson et al., 2021]. While such large-scale losses have not been recorded in the Baltic Sea, the region’s aquaculture sector is regularly exposed to the risk of HAB-related blooms. This is the case particularly through the contamination of shellfish with toxins produced by *Dinophysis* and *Alexandrium* species, which can cause serious illness in humans who consume affected mussels and oysters [Karlson et al., 2021]. Precautionary closures of mussel and oyster farming areas are now a regular part of coastal management in northern Europe, imposing costs on producers and disrupting supply chains even in years when no human illness actually occurs.

In the case of wild fisheries, the impact of the bloom is less clear but equally real. Recurring events of low oxygen in the bottom waters, caused in part by the decomposition of the bloom biomass, have had a growing effect on the habitats used by the commercially important fish species for feeding and reproduction. The long-term impact of this habitat degradation on the fish of the Baltic is hard to quantify from other factors such as over-fishing and changes in food availability due to climate change, but it is one more factor acting on heavily exploited species.

The economic effects of HABs on fishing communities in other areas provide a useful framework for considering the potential extent of the losses in the Baltic region. A detailed study of the economic effects of HABs on a recreational razor clam fishery in the US Pacific Northwest coast found that the loss of revenues from a single season’s closure due to HAB contamination could exceed 840,000 US dollars for the businesses of a single small community, and that this would extend beyond the fishing businesses to include gas stations, food stores, accommodation providers, and restaurants [Weir et al., 2022]. Perhaps more importantly, the study also found that the economic value of early warning systems, measured by what affected communities would be willing to pay to avoid closures, far exceeded the cost of the systems, providing a clear economic case for sustained investment in HAB surveillance [Weir et al., 2022]. While the figures involved in this study are not directly applicable to the Baltic region, the underlying principles are the same: the cost of HABs to fishing-dependent communities is serious number that is likely to rise with the frequency of the blooms.

5.3 Tourism, Recreation, and Coastal Communities

For the tens of millions of people who visit the the Baltic Sea every summer, algal blooms are not an abstract concern, but a tangible, even alarming, and certainly often maddening aspect of the vacation experience. The shores of the Baltic Sea, in Sweden, Finland, Germany, Poland, the Baltic states, and other locations, are home to a significant coastal tourism economy centered on swimming, sailing, fishing, and enjoying the natural scenery. When blooms arrive, they threaten all of these activities simultaneously, covering beaches in green-blue scum, closing bathing areas for public health reasons, and transforming the clear coastal waters that attract visitors into something that many describe as repellent.

Survey research carried out among tourists in the coastal areas of the Scania region in southern Sweden, an area that experienced severe and repeated bloom events between 2002 and 2006, offers detailed information about the ways in which tourists react to bloom events [Nilsson and Gössling, 2013]. The findings were surprising in a number of ways. For instance, the level of awareness of algal blooms among tourists in the coastal areas of the region was extremely high. 97.8% of the 3,217 survey respondents reported that they had heard of the phenomenon, which is an unusually high level of awareness of an environmental issue [Nilsson and Gössling, 2013]. The associations that tourists made with the blooms were overwhelmingly negative, with the most common associations involving threats to bathing, water quality, and general environmental concern.

When asked about their behavioral responses, a quarter of respondents admitted to changing their activities in response to the blooms, mainly through reducing or abandoning swimming. More significantly from an economic perspective, around 6% of all respondents reported either shortening their stay and moving to another area, or canceling their holiday altogether [Nilsson and Gössling, 2013]. Of the sample who had previously been experienced blooms but had decided not to return to a particular destination, nearly a third gave eutrophication as their primary reason for this decision, a factor that suggests a level of reputational damage to destinations experiencing blooms that extends far beyond the impact of any single summer season [Nilsson and Gössling, 2013].

These behavioural responses carry direct economic consequences that, while difficult to quantify precisely at the regional scale, are likely to be substantial given the size of the Baltic coastal tourism industry. The diversion of even a small percentage of the tourism population from bloom-affected destinations will have significant economic implications for the tourism sector, including accommodation providers, food outlets, retailers, and recreation operators – an economic loss that, unlike the economic loss from a single fishery closure, will occur quietly and it will accumulate over several tourism seasons. There is also the potential for longer-term economic damage to destinations that experience regular bloom events. These areas may struggle to attract first-time visitors, even during bloom-free periods, if the risk becomes embedded in the destination’s image.

Media plays an important and somewhat double-edged role in shaping these dynamics. On one hand, the coverage of the bloom events contributes to the overall awareness of the issue, allowing people to make informed choices about the location and time of their swimming activities. On the other hand, survey evidence suggests that media coverage sometimes amplifies the severity of risks associated with the bloom events, to the point that it even prevents people from visiting areas and time periods that were not affected by the blooms at all [Nilsson and Gössling, 2013].

5.4 Public Health Implications

Although the risk of health effects of Baltic algal bloom to the general public is certainly valid, it is also often misrepresented or overstated in public perception. Contact with algal bloom water, such as swimming through a bloom of algae floating on the surface, may cause skin and eye irritation, as well as gastrointestinal symptoms, particularly for children and sensitive individuals. More serious health risks arise from the consumption of shellfish or fish that have accumulated cyanobacterial toxins or other HAB-related compounds, which can cause liver damage or neurological effects depending on the toxin involved [Karlson et al., 2021]; [Karlson et al., 2022].

In the Baltic context, the toxin of greatest public health concern is nodularin, produced by *N. spumigena*, which has been detected in the tissues of commercially harvested fish species including herring, salmon, and cod, as well as in blue mussels [Karlson et al., 2022]. The monitoring of shellfish for HABs, which is now a legal requirement throughout the European Union, is an important safety net, and it is largely this monitoring activity that has prevented human illness resulting from Baltic bloom-related toxins in recent years. However, as bloom activity and extent increase, maintaining the coverage and responsiveness of these monitoring systems will become increasingly more difficult and expensive.

The public health aspect of Baltic bloom events also overlaps with the impact on tourism discussed above. The information provided to the public about health risks associated with blooms, whether by public health authorities, environmental protection organizations, or the mass media, has a direct impact on the actions of tourists and residents alike. This is not only a public health issue, but it is also an economic issue for the communities along the coast who rely on tourists to provide a large percentage of their annual income.

6 Monitoring, Detection, and Forecasting

The ability to observe the blooms, understand the circumstances that lead to their formation, and predict where and when they will occur is a critical prerequisite for responding effectively to the problems that they pose. This is, of course, much easier to state than to achieve in practice. Algal blooms are inherently unpredictable: they can appear and disappear within days, move in response to winds and currents, and vary dramatically in intensity from year to year for reasons that aren’t always clear from available data. Developing better monitoring and forecasting systems to anticipate blooms, or at least respond quickly enough to limit their impact, has become one of the major operational challenges facing the Baltic Sea region.

6.1 Traditional Monitoring: The Foundation of Long-Term Knowledge

The oldest and, in many ways, still the most scientifically valuable way of approaching the issue of algal bloom monitoring in the Baltic is the sampling of water and the microscopic investigation of phytoplankton communities. Long-term phytoplankton monitoring programmes, coordinated through the Helsinki Commission and national institutions such as the Swedish Meteorological and Hydrological Institute (SMHI), have been sampling water at fixed points throughout the Baltic for several decades, offering a record of the composition of phytoplankton communities, their size, and the time of year [Karlson et al., 2022]. This approach is irreplaceable in terms of offering a baseline against which more recent changes can be compared, and it is the foundation of much that is known about the long-term evolution of cyanobacterial communities, the movement of blooms northwards, and the relationships between nutrient conditions and the intensity of blooms.

The limitation of traditional monitoring, however, is rooted in the coverage. Sampling stations, visited monthly or, at best, weekly, may detect temporal trends in bloom dynamics at particular locations, but they will fail to detect the full spatial complexity of bloom dynamics in a sea as extensive as the Baltic. A bloom may form quickly between sampling visits, or it may form in a location where monitoring stations are absent, and may remain undetected until it reaches the coast and is noticed by the public. This limited data coverage across space and time is a persistent problem in Baltic bloom science, limiting the ability to validate forecast models and to provide the real-time information that coastal managers and the public need during bloom events [Karlson et al., 2022], [Roiha et al., 2010].

To complement fixed station monitoring, autonomous water quality monitoring systems installed on merchant vessels, known as FerryBox, have greatly improved the spatial extent of in-water monitoring. These systems, which continuously collect water parameters such as phycocyanin fluorescence, an indicator of cyanobacteria concentration, as vessels travel along fixed routes in the Baltic Sea, offer a spatial resolution that cannot be achieved with fixed station monitoring [Karlson et al., 2022]. Phycocyanin fluorescence data from FerryBox systems has been shown to correlate well with cyanobacteria biomass estimated by microscopy, making it a valuable complement to traditional sampling. However, like satellite monitoring, FerryBox monitoring is restricted to particular routes and cannot offer basin-scale monitoring of water quality.

6.2 Satellite Remote Sensing: Seeing the Bigger Picture

The most significant revolution in the monitoring of blooms in the Baltic over the past three decades has been the introduction of satellite remote sensing, where the surface accumulations of algal blooms over the whole sea are detected using instruments carried on satellites in space. Satellites carrying ocean color sensors, such as the NASA MODIS instruments carried by the Aqua and Terra satellites, detect the characteristic changes in the way the water reflects sunlight when there is a large amount of surface accumulations of the cyanobacteria. The location and intensity of the surface accumulations of the blooms over the whole sea are mapped at a resolution of about one kilometer each day, if the weather allows [Karlson et al., 2022].

The importance of satellite monitoring for the Baltic cannot be exaggerated. It has shown the spatial distribution of bloom location, which would otherwise be completely unknown from a ship-based survey program, verified the northward expansion of blooms into the Bothnian Sea, determined the years when blooms reached extreme extent, and provided the multidecadal record upon which much of the statistical analysis of bloom trends and environmental correlations discussed in the preceding sections of this review is built [Kahru et al., 2020]; [Kahru et al., 2025]; [Karlson et al., 2022]. To communicate information to the public, satellite-based maps of blooms are a vital tool, particularly to advise managers, tourists, and others about current bloom conditions in near real time.

Satellite monitoring does, however, possess certain inherent limitations that need to be understood for the effective use of the information obtained from satellite monitoring. The most basic limitation of satellite monitoring is that the satellite cannot measure the entire population of cyanobacteria in the water, as the satellite is essentially measuring only the accumulation of cyanobacteria on the water surface. They do not capture the broader cyanobacterial community distributed through the surface waters, which can be significant even when no surface accumulation is visible. This implies that the satellite and the microscopic methods of calculating the biomass of cyanobacteria are measuring different parameters, and the information obtained from the satellite needs to be interpreted properly [Karlson et al., 2022]. Satellite monitoring faces significant challenges in the Baltic region, where frequent cloud cover blocks observations and sediments in the water can interfere with optical signals. Despite these limitations, this approach remains the only

practical tool for comprehensive, basin-wide, near-real-time bloom monitoring across the full extent of the Baltic Sea.

6.3 Probabilistic Approaches to Bloom Forecasting

While it is valuable to know that a bloom is happening right now, it might be even more valuable to know that a bloom is likely to happen in the next two weeks. Bloom forecasting, or the prediction of when, where, and how a bloom might happen, represents the latest focus of operational Baltic Sea bloom science. Bloom forecasting is more challenging than weather forecasting, which has benefited from decades of model development and data collection, because it must account for biological processes that are naturally more variable and complex than atmospheric physics.

One promising method that has been shown to work in the Baltic is called ensemble forecasting. This approach, developed in meteorology, runs a computer model of the sea’s environment many times simultaneously, each time with slightly different weather conditions. Instead of generating one prediction, this creates a set of possible predictions that can quantify uncertainty and communicate it effectively to decision-makers [Roiha et al., 2010]. For example, it can produce probability maps showing the chance of bloom conditions exceeding a given threshold in different regions over the coming weeks.

A case study applying this approach in the Baltic Sea during summer 2008 showed promising results. The ensemble forecasts successfully captured the spatial distribution of bloom risk and the effects of upwelling events on bloom formation. While the predictions weren’t perfect, they were accurate enough to be useful for guiding environmental managers [Roiha et al., 2010]. However, the study also revealed a key limitation of this approach, which earlier research had also identified. The forecast models require measurements of phosphorus concentrations in the water during winter and spring as crucial input data. When this information is unavailable, it significantly reduces the accuracy of the forecasts. More broadly, validating bloom forecasts against real observations remains a major challenge because of the limited monitoring data available during summer bloom periods. Expanding automated monitoring networks like FerryBox systems and fixed stations could address this problem.

The ensemble method also has significant potential for application in risk communication and economic planning. By giving probability-based predictions instead of simple yes-or-no answers, this method helps decision-makers act proactively. Businesses, fisheries managers, and health authorities can use this information to close harvesting areas or issue swimming advisories before blooms develop [Roiha et al., 2010]; [Weir et al., 2022]. As mentioned in the previous section, the value of such advance warning can greatly exceed the cost of the systems that make such prediction possible.

6.4 Machine Learning Approaches to Bloom Prediction

In addition to physically based modeling techniques, researchers are increasingly exploring machine learning approaches for bloom prediction and detection. These techniques belong to the broader field of artificial intelligence, which unlike traditional models, machine learning algorithms learn patterns from the data rather than being explicitly programmed [Park et al., 2024]. Machine learning is appealing for bloom research because blooms involve complex interactions between many environmental factors, and these techniques excel at detecting complicated patterns in data.

Researchers have used many different machine learning approaches for bloom prediction in water systems around the world. Simple techniques like random forest and gradient boosting rely on decision trees to process large amounts of environmental data. More advanced methods use deep learning, including Long Short-Term Memory networks that can analyze patterns in time series data [Park et al., 2024]. Research on bloom prediction in various water bodies has demonstrated the effectiveness of machine learning approaches. These techniques consistently produce more accurate predictions than traditional statistical methods. The performance gap becomes even larger when researchers have access to extensive datasets [Park et al., 2024]. Coastal studies have shown that machine learning can predict blooms with 86% accuracy several days in advance. Achieving similar performance in the Baltic Sea would be extremely valuable.

A particularly promising recent development is explainable artificial intelligence. These techniques allow researchers to look inside machine learning models and identify which environmental variables drive their predictions [Park et al., 2024]. This is valuable because machine learning becomes a scientific discovery tool rather than just a prediction black box. These methods can reveal surprising insights, such as showing that weather conditions weeks before a bloom may be more important than nutrient levels at the time of blooming. Such insights have practical

applications for both monitoring and modeling. They can guide monitoring programs toward the most important variables and help scientists decide which factors to include in future mechanistic models.

For Baltic Sea bloom prediction specifically, machine learning hasn't been implemented on an operational scale, despite the clear potential. The Baltic Sea has some of the longest and most comprehensive phytoplankton monitoring records of any sea in the world. This is exactly the type of extensive dataset that machine learning techniques need to perform well. To make the most of machine learning opportunities, researchers need more than access to current datasets. They also need ongoing support for the monitoring programs that generate the comprehensive, multi-decade data that machine learning techniques require [Park et al., 2024]; [Karlson et al., 2022].

7 Conclusions

The Baltic Sea is, in many respects, a small-scale example of the broader environmental issues facing the world's coastal and marine environments. It is under pressure from many different directions. Decades of intensive farming have left a legacy of nutrient pollution, climate change is accelerating, and millions of people depend on the sea in ways that sometimes conflict with each other. Algal blooms are perhaps the most obvious symptom of all of this pressure, and this review has sought to pull together what is currently known about them and how they might be monitored and managed.

Several conclusions may be drawn with relative clarity from the evidence discussed above. First and foremost, algal bloom in the Baltic is not an issue that will go away on its own. The conditions that support algal blooms are deeply embedded in the Baltic environment. These include phosphorus buildup in seafloor sediments, warming and layering of the water, and the ability of cyanobacteria to produce their own nitrogen from the atmosphere. These conditions will persist for decades, regardless of how optimistic our management plans may be. The so-called vicious circle of eutrophication is not a temporary phase but a stable state from which the Baltic will require a strong effort to escape [Munkes et al., 2021]; [Viitasalo and Bonsdorff, 2022].

Second, the factors that control bloom formation operate over several scales, which have important implications for both science and management. The long-term intensity of blooms depends mainly on phosphorus availability and bottom-water oxygen depletion. Fortunately, these are the factors that nutrient reduction programs can directly address [Kahru et al., 2020]. At the annual scale, whether blooms actually occur in any given year depends primarily on weather conditions. Specifically, the combination of sunlight and wind patterns during midsummer determines bloom development, and these factors are beyond management control [Kahru et al., 2020]; [Kahru et al., 2025].

Third, the impacts of algal blooms in the Baltics are more extensive, far-reaching, and multi-dimensional than might be recognized within a purely scientific context. These impacts begin with the destruction of coastal animal communities and the contamination of food webs with liver toxins [Gubelit and Berezina, 2010]; [Karlson et al., 2022]. They continue to spread through the disruption of fishing and aquaculture industries throughout northern Europe [Karlson et al., 2021]; [Weir et al., 2022], and tourists avoiding bloom-affected destinations, creating long-term economic damage for the Baltic regions [Nilsson and Gössling, 2013]. The overall economic cost of algal blooms in the Baltics, across all these sectors, has yet to be fully quantified, but the available data strongly suggests it is considerable, and growing.

Fourth, while important progress has been made in the monitoring and forecasting of Baltic Sea blooms, important gaps and limitations remain. Satellite remote sensing has revolutionized the spatial extent of bloom observations, but it cannot yet monitor subsurface bloom activity or see through cloud cover [Karlson et al., 2022]. Traditional monitoring approaches have generated valuable long-term records, but they don't provide enough data coverage across different locations and times to support quick management decisions. While ensemble forecasting techniques show great potential for seasonal bloom prediction, their accuracy is constrained by limited observational data needed to validate the models [Roiha et al., 2010]. Machine learning offers another promising avenue, demonstrating success in bloom forecasting for other aquatic systems, though it hasn't been widely applied to the Baltic [Park et al., 2024]. Emerging molecular monitoring techniques can track toxic cyanobacteria from rivers to open ocean, but like machine learning approaches, these methods remain largely untested in Baltic waters.

Fifth, climate change is not something that will happen in the future and can be dealt with once the eutrophication problem has been solved. Climate change is an agent that is already affecting the environment in ways that will make the management of the Baltic blooms an increasingly

difficult problem [Viitasalo and Bonsdorff, 2022]; [Karlson et al., 2022]. The Baltic Sea is predicted to become fresher under climate change, adding another complication to bloom management. Some bloom-forming species will benefit from this change while others will suffer, making it genuinely uncertain how the problem will evolve in the future [Tulatz et al., 2025].

Against this challenging background, however, the review also suggests that there are grounds for some optimism based on couple of different reasons. First, international nutrient management strategies have clearly reduced external nutrient loading in the Baltic over the past decades, despite imperfect implementation. Modeling studies consistently suggest that continued and enhanced application of these strategies, particularly the Helsinki Commission’s Baltic Sea Action Plan, could offset climate change effects on bloom intensification if executed properly and given sufficient time [Viitasalo and Bonsdorff, 2022]; [Munkes et al., 2021]. Scientific tools for monitoring, understanding, and predicting blooms are advancing quickly. Baltic Sea monitoring programs have created exceptional long-term datasets that, when combined with modern machine learning and molecular techniques, could enable the development of powerful new predictive capabilities [Park et al., 2024]; [Karlson et al., 2022].

This review identifies several priority areas where targeted research and monitoring efforts would most effectively advance bloom science. For example, better understanding of biological processes like species competition could significantly improve bloom predictions. Such processes drive a lot of the unexplained year-to-year variation in bloom intensity, and understanding them would enhance both mechanistic and machine learning models. The Bothnian Sea also needs more monitoring infrastructure, as bloom patterns are changing significantly there but current monitoring is limited. Developing genetic monitoring approaches to track toxic cyanobacteria connections between Baltic freshwater systems and coastal waters could open up new possibilities for early warning systems. Additionally, integrating existing monitoring data from satellites, FerryBox systems, fixed stations, and molecular techniques into a coordinated, internationally-managed climate service could transform how the Baltic Sea region anticipates, communicates about, and responds to bloom challenges [Karlson et al., 2022].

The Baltic Sea has long been described as a system in crisis, and when it comes to algal blooms, this description fits. However, the Baltic is also intensively studied, closely monitored, and surrounded by countries with both the institutional capacity and collective commitment to tackle the problem. The science reviewed here provides a clear picture of the work ahead and the tools available to address blooms. The key question is whether these tools will be used with sufficient ambition and cooperation to move beyond the current struggle with summer blooms in the coming decades.

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